

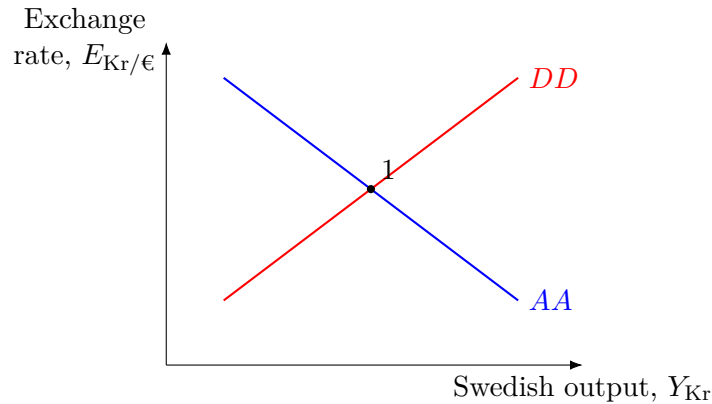
Problem Set 9

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Problem 1.

- (a) The AA-DD model for Sweden, using $E_{\text{Kr}/\text{€}}$ on the vertical axis and Swedish output on the horizontal axis, is:



- (b) A **permanent decrease** in the Swedish money supply is contractionary monetary policy. In the short run, prices are fixed, so

$$\frac{MS_{\text{Kr}}}{P_{\text{SE}}} \downarrow \Rightarrow R_{\text{Kr}} \uparrow .$$

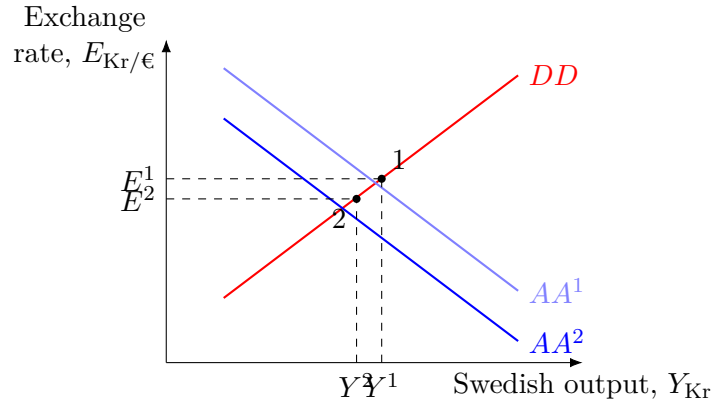
This raises the return on Swedish deposits and causes the krona to appreciate:

$$E_{\text{Kr}/\text{€}} \downarrow .$$

Because the change in money supply is permanent, people also expect a stronger krona in the future, so

$$E_{\text{Kr}/\text{€}}^e \downarrow .$$

That strengthens the appreciation effect even more. Therefore, the *AA* curve shifts down, and because the policy is permanent it shifts down by more than it would under a temporary contraction. The *DD* curve does not shift.



So, in the short run,

$$Y_{Kr} \downarrow, \quad E_{Kr/\epsilon} \downarrow.$$

Swedish output falls, and the krona appreciates.

(c) In the short run, goods prices are assumed to be sticky.

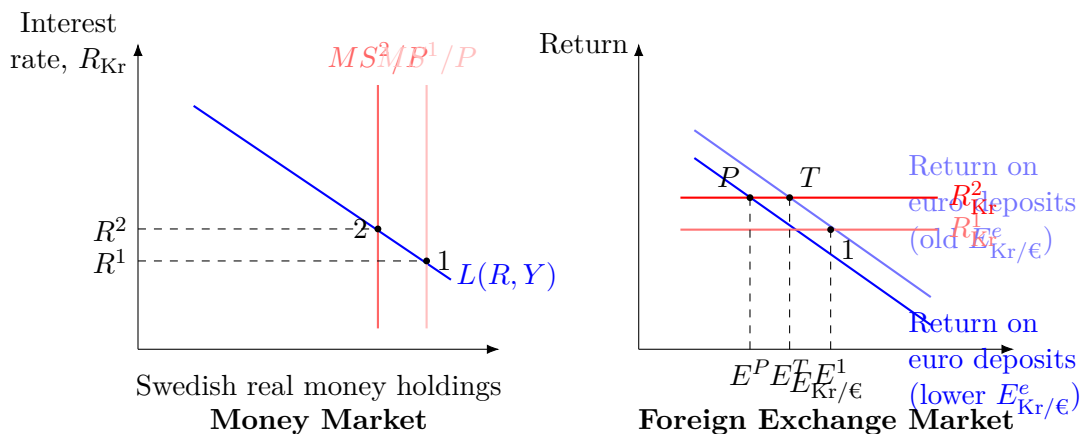
- (i) The price in kronor of a bag of frozen Swedish meatballs sold in Stockholm is unchanged in the short run. This is exactly the short-run fixed-price assumption.
- (ii) The price in euros of a bag of Swedish meatballs sold in Paris rises. If the Swedish price in Stockholm is fixed in kronor, then the euro price is

$$P_{\epsilon}^{\text{Paris}} = \frac{P_{Kr}^{\text{Stockholm}}}{E_{Kr/\epsilon}}.$$

Since the krona appreciates, $E_{Kr/\epsilon}$ falls. Therefore the denominator falls, so the euro price rises.

(d) If the money-supply reduction is temporary instead of permanent, the change in prices from part (c) is:

- (i) The same for the Stockholm price in kronor: it is still unchanged in the short run.
- (ii) Smaller for the Paris price in euros. The euro price still rises, but it rises by less than in part (c).



A temporary decrease in the money supply raises Swedish interest rates, so the return on kronor deposits rises from R_{Kr}^1 to R_{Kr}^2 . This moves the equilibrium exchange rate from E^1 to E^T , so the krona appreciates.

But a permanent decrease in the money supply does more than that. It also makes people expect a stronger krona in the future, so $E_{\text{Kr}/\text{€}}^e$ falls. That shifts the return on euro deposits downward and moves the exchange rate further to E^P . Therefore,

$$E^P < E^T < E^1.$$

So the temporary contraction causes a smaller appreciation of the krona than the permanent contraction. Therefore the rise in the Paris euro price is smaller under a temporary decrease in the money supply.

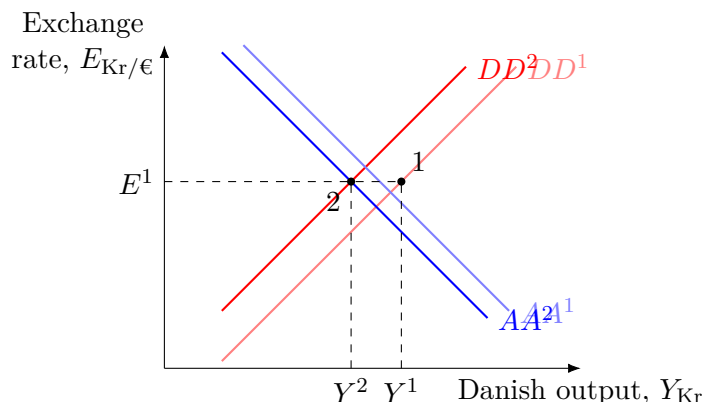
Problem 2.

- (a) If people expect the Danish central bank to permanently increase the money supply in the near future, then they will expect higher prices in the future. In the long run, a higher money supply implies a higher domestic price level.
- (b) Now assume those inflation expectations cause prices to rise today. In the AA-DD model:

$$P_{\text{DK}} \uparrow \Rightarrow \frac{MS_{\text{Kr}}}{P_{\text{DK}}} \downarrow \Rightarrow R_{\text{Kr}} \uparrow,$$

so the AA curve shifts down.

Also, when Danish prices rise, Danish goods become more expensive relative to foreign goods, so the current account falls. Therefore the DD curve shifts left.



The economy moves from point 1 to point 2. Output falls:

$$Y_{\text{Kr}} \downarrow.$$

$$E_{\text{Kr}/\text{€}} \text{ unchanged.}$$

- (c) Since output falls below full employment, the Danish central bank should undertake expansionary monetary policy, meaning it should increase the money supply. That shifts the AA curve upward, lowers interest rates, and depreciates the krone enough to move the economy back to full employment.

(d) Yes, in this setup it is a self-fulfilling prophecy.

The original belief is that the central bank will increase the money supply in the future. That belief causes people to expect higher future prices. Those expectations then raise prices today. The higher current price level reduces real money balances and hurts the current account, which lowers output. Because the central bank has a full-employment mandate, it now has an incentive to increase the money supply to bring output back up. So the original expectation can cause behavior that makes the monetary expansion actually occur.